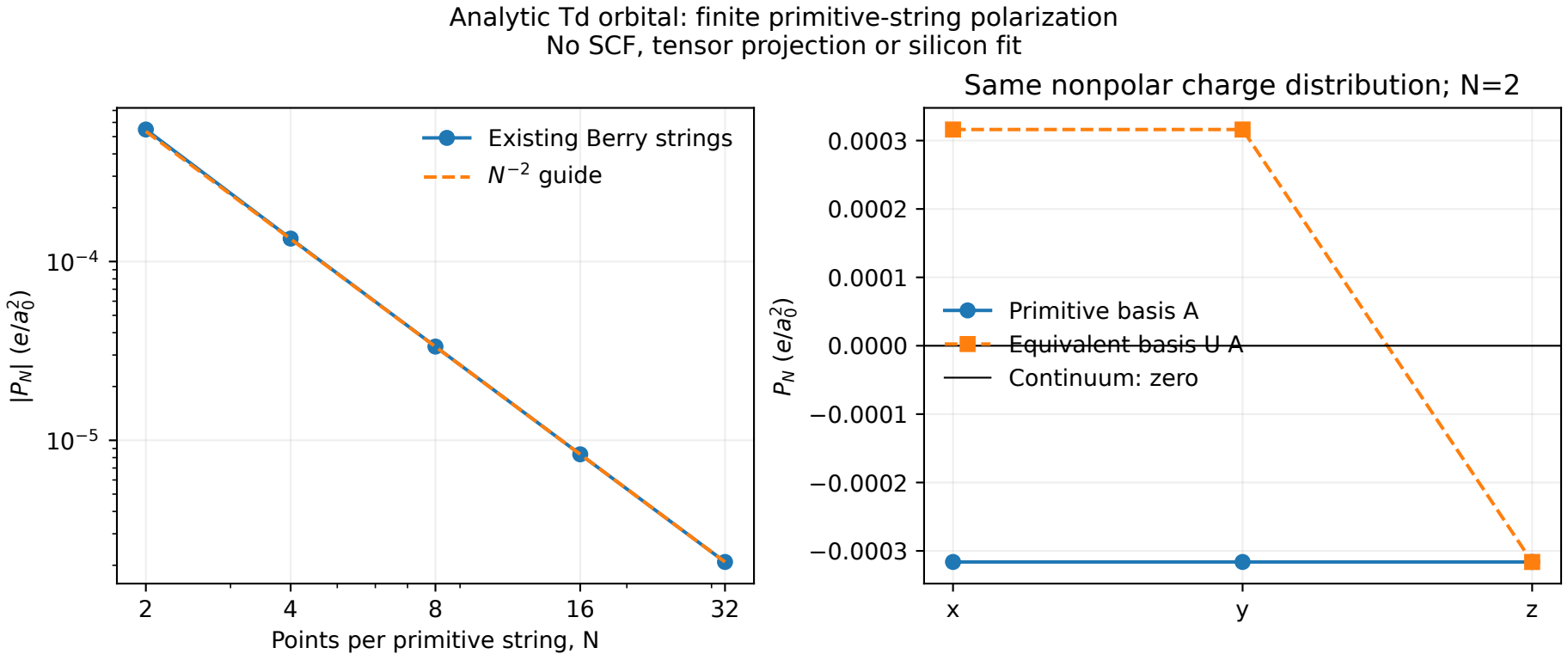
Analytic Td orbital: finite primitive-string polarization
No SCF, tensor projection or silicon fit
Existing Berry strings
N^{-2} guide
$|P_N|$ (e/a_0^2)
10^{-4}
10^{-5}
2
4
8
16
32
Points per primitive string, N
Same nonpolar charge distribution; N=2
P_N (e/a_0^2)
0.0003
0.0002
0.0001
0.0000
−0.0001
−0.0002
−0.0003
Primitive basis A
Equivalent basis U A
Continuum: zero
x
y
z